Psychonomic Bulletin & Review
https://doi.org/10.3758/s13423-025-02655-z

BRIEF REPORT

Global precedence effects account for individual differences in taxonomic and thematic relations recognition performance

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Accepted: 25 January 2025

Abstract

This study examined whether individuals with higher levels of global precedence recognized thematic relations faster and exhibited a stronger preference for them. In Study 1, the Global–Local Precedence Index was calculated based on results from the Navon task to reflect the degree of individual global precedence. Preferences for thematic or taxonomic relations and recognition speed were measured using the forced-choice triad task and the similarity-matching task, respectively. The results showed that participants with a higher Global–Local Precedence Index had a higher proportion of thematic choices in the forced-choice triad task and identified thematically similar options faster in the similarity-matching task. In Study 2, we replaced the stimulus set of the forced-choice triad task with a larger and more diverse set of stimuli and replaced the similarity-matching task (a word-based task) with a semantic priming task (a picture-based task). The results still showed that participants with a higher Global–Local Precedence Index chose thematic options more frequently in the forced-choice triad task and responded faster to correctly judge whether the orientation of Gabor patches was the same or different after thematic priming. The findings indicate that individuals with higher global precedence recognize thematic relations faster and show a stronger preference for them.

Keywords Semantic memory · Thematic relations · Global precedence effect · Individual differences

Conceptual knowledge underpins virtually all human cognitive processing. It enables us to recognize language, know how to handle objects, allows us to predict interactions among different objects in the world, and recall past events and imagine future ones (McRae & Jones, 2013; Metusalem et al., 2012; Rogers et al., 2004). Contemporary cognitive psychology research indicates that conceptual knowledge in semantic memory is primarily organized in two different ways: taxonomic relations and thematic relations (Mirman et al., 2017). Taxonomic relations are defined by membership in a common category based on shared features, such as biological taxonomic hierarchies, or by examining the similarity structure that emerges from participants listing similar features (Kalénine & Buxbaum, 2016; Mirman et al., 2017). Thematic relations are the external or complementary relations among objects, events, people, and other entities that co-occur or interact together in space, time, events, or situations (Lin & Murphy, 2001; Mirman et al., 2017). Such relations can include relations that are explicitly tied to specific roles in events or schemas (e.g., Pluciennicka et al., 2016; Wamain et al., 2015).

Researchers have examined distinctions between taxonomic and thematic relations from multiple perspectives, such as how both taxonomic and thematic relations independently contribute to semantic relatedness (e.g., Estes et al., 2012) and through evidence from neuropsychological double dissociation (e.g., Merck et al., 2014). In addition, another area of research highlights differences among individuals in the relative strength with which they process thematic versus taxonomic relations (Estes et al., 2011; for a review, see Mirman et al., 2017). First, individuals exhibit distinct preferences for thematic versus taxonomic relations. In an early study by Gentner and Brem (1999), researchers, using the forced-choice triad task,[1] found that while many participants

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[1] In a forced-choice triad task, participants were presented with a reference word (e.g., "dog") along with two additional items: one taxonomically related (e.g., "cat") and the other thematically related (e.g.,

did not exhibit a consistent semantic preference in the task, some consistently demonstrated a thematic preference, while others consistently showed a taxonomic preference. Subsequent research has replicated these findings under a range of task instructions (Lin & Murphy, 2001), stimulus materials (Honke et al., 2020), and task types (Mirman & Graziano, 2012b). Second, the time course of recognizing thematic versus taxonomic relations varies among individuals. A recent study by Nah and Geng (2022), using a semantic priming task[2] that manipulated the types of semantic priming, found that individuals recognized thematic relations faster than taxonomic relations. Furthermore, studies using the visual world paradigm (Kalénine et al., 2012) and comparing comprehension times for word pairs (Estes, 2003) have similarly found faster thematic processing. Conversely, some studies have found that taxonomic relations are sometimes processed more rapidly (Honke et al., 2020). The time course of activation may be influenced by systematic preferences for thematic versus taxonomic relations.

What are the potential drivers of individual differences in thematic and taxonomic processing? Existing studies have identified factors such as developmental age, domain expertise and culture, and level of construal. Research has found that children and older adults (Berger & Aguerra, 2010; Brooks et al., 2014), East Asians (Ji et al., 2004), domain experts (Overcast et al., 1975; Sharp et al., 1979), and individuals with high-level construal (Li et al., 2023, 2024) all show a stronger preference for or greater sensitivity to thematic relationships, whereas they exhibit the opposite tendency for taxonomic relationships. However, in efforts to fully recognize the cognitive functions of the brain, these factors from different perspectives should be regarded as potentially useful information. They may provide deeper insights into the underlying causes of these differences, thus contributing to a more comprehensive recognition of the brain's cognitive structure (Gerlach & Poirel, 2018; Gerlach & Starrfelt, 2018; Wilmer, 2008).

Among the aforementioned factors, age, cultural background, domain expertise, and level of construal all appear to be closely related to individual differences in perceptual processing. Previous studies have found that children and older adults (Bouhassoun et al., 2022; Krakowski et al., 2016), domain experts (Lewis & Dawkins, 2015), East Asians (Boduroglu et al., 2009), and individuals with high-level construal (Li et al., 2024) exhibit a stronger tendency toward global precedence when processing information. More importantly, a growing body of research indicates that cognitive activities involving thematic relationship representations rely heavily on the temporo-parietal cortex (TPC; e.g., Kalénine & Buxbaum, 2016; Kalénine et al., 2009; Mirman & Graziano, 2012a). Studies have shown that the TPC plays a critical role in integrating spatially distributed objects into a single coherent percept, a process often referred to as "Gestalt perception" or global processing (Huberle & Karnath, 2012; Lestou et al., 2014). Therefore, we hypothesize that individual differences in the level of global precedence (the global precedence effect) may be one of the more direct driving factors underlying individual variability in thematic and taxonomic relationship representations. However, no studies have directly explored this possibility to date.

The global precedence effect refers to the phenomenon where the global aspects of a scene are processed more rapidly than its local details (Kimchi, 1992; Navon, 1977; Rezvani et al., 2020). Research suggests that individual differences in global precedence can account for a significant proportion of variance in tasks involving object differentiation. For instance, it has been found that the degree of global precedence is closely associated with performance in distinguishing faces from objects (Gerlach & Starrfelt, 2018). In the study by Gerlach and Starrfelt (2018), participants completed the Navon task, in which they were sequentially presented with large letters (global level) composed of smaller letters (local level). These letters could either be congruent or incongruent. Participants then performed a series of tasks measuring their ability to differentiate and recognize faces and objects. After completing all tasks, the researchers calculated the Global–Local Precedence Index.[3] Higher scores on this index indicate a stronger global processing preference. The results revealed that individuals with higher global precedence scores performed better in correctly identifying and categorizing both objects and faces. Moreover, the degree of global precedence was also found

Footnote 1 (continued)

"bone"). Participants were asked to judge which item was more similar to the reference word (`Gentner & Brem, 1999).

[2] In semantic priming task, participants were presented with pairs of objects that were thematically related, taxonomically related, or unrelated; subsequently, vertical Gabor patches appeared over each object. These Gabor patches were tilted 45° to the left or to the right, and participants were required to report whether the orientation of the two patches was the same or different. The results showed that participants processed Gabor patches on thematically related objects significantly faster than those on taxonomically related objects (`Nah & Geng, 2022).

[3] As demonstrated by `Gerlach and Krumborg (2014), many formerly used indexes derived from Navon's paradigm have probably been unreliable. To avoid this problem, Gerlach and Starrfelt (2018) use an index of global/local shape bias which is based on the standardized mean difference (Cohen's *d*) between RTs to Local Consistent and Global Consistent trials. In comparison with other indexes derived from Navon's paradigm, this index, which Gerlach et al. term the Global–Local precedence index, is pure, because it measures differences in global and local processing that are not confounded by interference effects (`Gerlach & Krumborg, 2014). The higher the score on this index, the faster are responses to global as compared with local shape characteristics.

to be strongly related to performance in cognitive tasks that involved distinguishing semantic or object-related information, such as deciding whether pictures represented real objects or nonobjects, and determining whether stimuli depicted natural objects or artifacts (Gerlach & Starrfelt, 2018). This raises an intriguing question: Could the global precedence effect also be related to performance in differentiating and representing various semantic relationships? More specifically, do individuals with higher levels of global precedence recognize thematic relations faster and exhibit a stronger preference for them?

First, research has shown that thematic relations can guide our expectations about what we are likely to encounter in scenarios or events (Mirman et al., 2017; Zhai et al., 2025). In the cognitive process of "expectation," both understanding the relationships between entities and predicting the dynamic development of their relationships rely more on the integration of these relationships and the activation of a global congruent context (Haddad et al., 2024; Maldei et al., 2020). Second, studies also suggest that thematic relations are more likely to depend on how the features of objects, such as their positions and functions, interact or relate within a scenario or event (e.g., Mirman et al., 2017). For example, Kalénine and Buxbaum (2016) found that thematic semantics are related to knowledge of object-use actions. Furthermore, prior research has indicated that individuals with a higher tendency for global processing prioritize the integration of objects based on their overall relations more quickly and efficiently (Li et al., 2019, 2024). Based on this, the present study hypothesizes that individuals with higher levels of global precedence will recognize thematic relations more quickly and show a stronger preference for them.

This study measured individuals' Global–Local Precedence Index using the Navon task (e.g., Gerlach & Starrfelt, 2018). Study 1 assessed participants' preferences for thematic or taxonomic relationships and their comprehension speeds through the forced-choice triad task (e.g., Li et al., 2023, 2024) and the similarity-matching task (e.g., Simmons & Estes, 2008). In Study 2, the stimulus set of the forced-choice triad task was expanded to include a larger and more diverse set of stimuli (Honke et al., 2020). Additionally, the similarity-matching task (a word-based task) was replaced by a semantic priming task (a picture-based task).

Study 1

Methods

Participants and design

In this study, we conducted an a priori sample size analysis using G*Power 3.1 to determine the necessary number of participants for our Pearson correlation analysis. We aimed for a medium effect size of 0.3, in line with Cohen's (1988) classification. With an alpha level set at 0.05 and a targeting power of 0.8, G*Power indicated that a sample size of approximately 68 participants was required to adequately detect a significant correlation (Faul et al., 2009). To enhance statistical power, a total of 112 university students willingly participated in the experiment, comprising 58 women and 54 men, with an average age of 21.57 years. Handedness was assessed using the Oldfield Edinburgh Handedness Inventory, and all participants were confirmed to be right-handed (Oldfield, 1971). The native language of all participants was Chinese, but each had been learning English since at least the age of 8 for a minimum duration of 12 years, enabling them to read the Latin alphabet fluently. All participants were from urban areas. Additionally, they were informed about the study, provided written informed consent, and received course credits upon completion of the experiment. The study was conducted following the Helsinki Declaration of 1964 and was approved by the Ethical Committee of Northwest Normal University (Reference: psynwnu-003). No participants were excluded from the analyses reported below.

Materials and procedure

Navon task This task aims to measure an individual's Global–Local Precedence Index. A higher index indicates a stronger degree of global precedence in the individual. In this task, participants were presented with a sequence of large letters, composed of smaller letters, on a computer screen. The relationship between the large letters and the small letters forming them varied; they could either be consistent (e.g., a large *H* made up of small *H*s) or inconsistent (e.g., a large *H* made up of small *S*s). The experimental design comprised four blocks presented in an ABBA sequence. Participants were tasked with identifying whether the large letter was an *H* or an *S* in two blocks and whether the small letter was an *H* or an *S* in the other two blocks. Each trial commenced with a 1,000 ms cross-shaped gaze point displayed initially at the center of the screen, followed by a 180-ms letter stimulus, and concluded with a 2,000-ms blank screen. Before the formal experiment, participants underwent a practice session of 10 trials. Only those who achieved 100% accuracy during the practice were permitted to proceed to the formal task. The stimulus materials and task procedures were adapted from Gerlach & Starrfelt (2018). Half of the participants began with global identity judgments. The large letters measured $3.91^\circ \times 5.25^\circ$ of visual angle, while the small letters measured $0.47^\circ \times 0.67^\circ$. The fixation cross, presented before the stimulus onset, subtended $0.95^\circ \times 0.95^\circ$ of visual angle. All stimuli were black and displayed on a white background on a computer screen.

Each block consisted of 40 trials, with 20 trials featuring consistent large and small letters and 20 trials featuring inconsistent letters. Stimuli were presented at four positions (top, bottom, left, and right) relative to the fixation cross. Within each block, an equal number of consistent and inconsistent stimuli (10 consistent/10 inconsistent) were shown at these four positions, with the center of the global shape positioned at 3.34° of visual angle from the fixation cross.

The forced-choice triad task This task aims to measure the degree of preference individuals show for thematic or taxonomic relations. The proportion of thematic matches selected by participants served as an index of their processing preference. A higher proportion of thematic matches indicates a stronger preference for processing thematic relations. In this task, 40 triads were sequentially presented on a computer screen, with their presentation order randomly determined by the computer. Each triad comprised a reference word, such as *lamp*, along with two comparative words. One comparative word served as a taxonomic match, such as *flashlight*, highlighting shared attribute characteristics, like luminous lighting. The other comparative word represented a thematic match, such as *desk*, indicating a complementary role in a thematic context, such as "learning" or "conducting official business" (see Fig. 1). Participants were instructed to quickly and accurately determine which option was most similar or related to the reference word. Half of the thematic match words appeared on the left side of the triads, while the other half appeared on the right. The 40 triads were from a prior norming study (Landrigan & Mirman, 2018). To ensure cross-cultural validity, the research methodology suggested by Ember and Ember (2001) was adopted, and the word materials underwent translation and back-translation processes to create the Chinese version used in the experiment. Additionally, following the evaluation method proposed by Landrigan and Mirman (2016), the Chinese version was normed by 21 participants through separate surveys. These surveys evaluated the taxonomic similarity (defined as sharing similar features or belonging to the same category; rated on a scale from 1 = *not similar at all* to 7 = *very similar*) and thematic relatedness (defined as co-occurrence in common events; rated on a scale from 1 = *not related at all* to 7 = *very related*) of the pairs. Finally, a difference score was calculated for each word pair. For thematic relations, a difference score by subtracting the taxonomic similarity rating from the thematic relatedness; For taxonomic relations, a difference score by subtracting the thematic relatedness rating from the taxonomic similarity rating. A higher difference score indicates that the thematic relatedness or taxonomic similarity of the word pair is more distinct or normative. Conversely, a difference score closer to 0 suggests that the word pair lacks a clear association in either thematic relatedness or taxonomic similarity. Results indicated that all translated word pairs maintained strong normative properties in either thematic relatedness or taxonomic similarity (for more information, see Supplemental Information Table 1).

Fig. 1 The example of the forced-choice triad items

The similarity-matching task This task aims to measure the speed at which individuals understand thematic or taxonomic relations. A faster correct response time indicates a quicker understanding speed. In this task, 60 triads were sequentially presented on a computer screen, with their presentation order randomly generated by the computer. Unlike the forced-choice triad task, each triad consisted of a reference word (e.g., *pencil*), a related option (either a thematic option, such as *eraser*, or a taxonomic option, such as *pen*), and an unrelated option (e.g., *broom*). Among the 60 triads, 30 included thematic-related options, and the other 30 included taxonomic-related options. Participants were instructed to quickly and accurately determine which option was most similar or related to the reference word. Half of the related options appeared on the left side of the triads, while the other half appeared on the right. The 60 triads were from prior studies (de Zubicaray et al., 2013; Mirman & Graziano, 2012a, 2012b). Similarly, to ensure the cross-cultural validity of the experiment, the research methodology suggested by Ember and Ember (2001) was employed, and the word materials underwent translation and back-translation to create the Chinese version used in the experiment. Additionally, as with the forced-choice triad task, the Chinese version was normed by 19 new participants through separate surveys. Results indicated that all translated word pairs maintained strong normative properties in terms of thematic relatedness or taxonomic similarity (for more information, see Supplemental Information Table 2).

In the study, to minimize the influence of task order effects on the results, participants were divided into six groups.

Each group completed the Navon task, the forced-choice triad task, and the similarity matching (recognition) task in a different sequence. For example, the first group completed the Navon task, followed by the forced-choice triad task, and then the similarity matching (recognition) task. The second group completed the tasks in the order of the forced-choice triad task, the Navon task, and the similarity-matching task. This pattern continued for the remaining groups. All tasks were implemented using the PsychoPy (Peirce et al., 2019).

Results and discussion

Paired-sample t tests and correlation analyses in the study were conducted using the JASP statistics program (Wagenmakers et al., 2018). This study was not preregistered, and all data associated with this article are available on the Open Science Framework (OSF: osf.io/2yu9g).

First, for each participant, we analyzed their processing inclination using binomial tests based on the number of thematic response selections made in 40 triads. The process resulted in 31 (27.7%) taxonomic biased responders, 29 (25.9%) thematic biased responders, and 52 (46.4%) ambiguous responders (see Table 1). Next, we calculated the average reaction time (ms) for participants to correctly identify the option (thematically related or taxonomically related) associated with the reference word in the similarity-matching task (see Table 1). A paired-sample t test revealed that participants responded significantly faster when selecting the thematically related option (M = 1482.126, SD = 81.491) compared with the taxonomically related option (M = 1558.285, SD = 131.211), $t(111) = -4.740$, $p < .001$, 95% CI [−107.994, −44.323], Cohen's $d = -0.448$.

Second, we calculated the average reaction time (ms) for participants to correctly respond during the Navon task (see Table 1). A paired-sample t test revealed that participants identified Navon global letters (M = 537.968, SD = 113.008) significantly faster than Navon local letters (M = 574.202, SD = 110.286), $t(111) = -2.655$, $p < .01$, 95% CI [−63.280, −9.189], Cohen's $d = -0.251$. This result aligns with findings from previous studies (e.g., Li et al., 2024; Navon, 1977). Additionally, following the methodology of prior research, we calculated the Global-Local Precedence Index for all participants, which is based on the standardized mean difference (Cohen's d) between reaction times for Local Consistent and Global Consistent trials. The higher the score on this index, the faster responses to global as compared with local shape characteristics. More importantly, correlational analysis revealed that the Global-Local Precedence Index significantly negatively correlated with the reaction speed of participants identifying thematically related options in the similarity-matching task, $r = -.563$, $p < .001$, 95% CI [−0.677, −0.421], and significantly positively correlated with the reaction speed of identifying taxonomically related options, $r = -.537$, $p < .001$, 95% CI [0.391, 0.657] (see Fig. 2).

In summary, the findings indicate that Individuals with higher levels of global precedence recognized thematic relations faster and exhibited a stronger preference for them.

Study 2

Although the results of Study 1 suggest that individuals with higher levels of global precedence recognized thematic relations faster and exhibited a stronger preference for them, existing research indicates that similarities between tasks can influence cognitive performance (Li et al., 2023). In Study 1, despite differences in the task demands and materials of the forced-choice triad task and the similarity matching (recognition) task, both were presented in a triadic format, which could have influenced the results. Therefore, in Study 2, we replaced the similarity-matching task (a word task) with a semantic priming task (a picture task). Additionally,

Table 1 Participants' performance under different task conditions in Study 1

Navon task			
Condition	RT (ms)		Acc (%)
Local Consistent	537.97 (113.01)		92.48 (6.97)
Global Consistent	574.20 (110.29)		90.23 (7.81)
The forced-choice triad task (40 triads)			
Thematic matching proportion mean (%)	Thematic-biased responder (thematic response > 25)	Taxonomic-biased responder (thematic response < 15)	Ambiguous responders
48.17 (23.11)	29 (25.9%)	31 (27.7%)	52 (46.4%)
The similarity-matching task			
Semantic similarity type	RT (ms)		Acc (%)
Thematic	1,482.13 (81.50)		91.35 (8.99)
Taxonomic	1,558.29 (131.21)		89.21 (7.65)

in Study 2, we replaced the stimulus set of the forced-choice triad task into a much larger stimulus set (Honke et al., 2020; Shi & Li, 2024).

Methods

Participants and design

Consistent with Study 1, to enhance statistical power, 113 university students willingly participated in the experiment, comprising 62 women and 51 men, with an average age of 21.47 years. Handedness was verified according to the Oldfield Edinburgh test; all participants were right-handed (Oldfield, 1971). Similarly, all participants were native speakers of Chinese, but each had been learning English since at least the age of 8 and had continued for at least 12 years, enabling them to read the Latin alphabet without difficulty. All participants were from urban areas. Furthermore, all were informed about and signed a consent form, and they received course credits upon completion of the experiment. The study was conducted following the Helsinki Declaration of 1964 and was approved by the Ethical Committee of Northwest Normal University (Reference psynwnu-003). No participants were excluded from the analyses reported below.

Materials and procedure

Navon task The Navon task was consistent with that of Study 1.

The forced-choice triad task The task requirements for the forced-choice triad task were consistent with those in Study 1. However, unlike Study 1, the forced-choice triad task in Study 2 included 80 triads (word triads). These 80 triads were adapted and standardized by Honke et al. (2020), while the translated and standardized Chinese version was derived from the work of Shi and Li (2024). However, since we could not find a report from Shi and Li (2024) reevaluating the similarity of the words in this version, we followed the method of Landrigan and Mirman (2018) and additionally recruited 20 participants to conduct a standardized evaluation of the semantic similarity between the words (for more information, see Supplemental Information Table 3).

The semantic priming task This task also aims to measure the speed at which individuals understand thematic or taxonomic relations. A faster correct response time following thematic or taxonomic semantic activation indicates a quicker understanding speed. The task comprised a total of 600 trials, with 200 trials each designated for thematic priming, taxonomic priming, and neutral priming, respectively. As illustrated in Fig. 4, each trial began with a 500-ms fixation point displayed at the center of the computer screen. Immediately afterward, two objects appeared on either side of the fixation point, enclosed by borders that indicated their relational context: thematically related (blue border), taxonomically related (red border), or neutral (green border). This display persisted for 750 ms. Subsequently, two vertical Gabor patches were superimposed at the center of each object, tilted either 45° to the left or 45° to the right, and presented for a duration of 200 ms. Participants were instructed to respond as quickly and accurately as possible, indicating whether the orientations of the Gabor patches were the same or different by pressing the corresponding keys. After the Gabor patches disappeared, participants were allotted 2,000 ms to provide their response. Notably, the semantic relationships between the objects were task-irrelevant and bore no predictive value for the orientation of the Gabor patches. Similar to previous studies, there were 10 sets of thematically, taxonomically, and neutrally related images. These images and the experimental procedures were drawn from a prior norming study (Nah & Geng, 2022). Nah and Geng (2022) rigorously standardized the image features and the semantic similarity of all picture stimuli (see Supplemental Information or the materials section of the Nah & Geng, 2022, study). Considering cross-cultural applicability, we also recruited an additional 20 participants to rate the semantic relevance of the stimuli, with results indicating that all picture stimuli maintained good standardization in thematic relatedness or taxonomic similarity (for more information, see Supplemental Information).

As in Study 1, to minimize the influence of task order effects on the results, participants were divided into six groups. Each group completed the three tasks in a different order.

Results and discussion

First, consistent with Study 1, the results of binomial tests indicated that 36 (31.9%) taxonomic biased responders, 26 (23.0%) thematic biased responders, and 51 (45.1%) ambiguous responders (see Table 2). Next, we calculated the average reaction time (ms) of participants in a semantic priming task when correctly judging whether the orientation of Gabor patches was the same or different following thematic and taxonomic priming (see Table 2). A paired-sample t test revealed that participants responded significantly faster after thematic priming ($M = 587.339$, $SD = 74.088$) than after taxonomic priming ($M = 626.057$, $SD = 81.781$), $t(112) = -3.499$, $p < 0.001$, 95% CI [-60.646, -16.791], Cohen's $d = -0.329$.

Table 2 Participants' performance under different task conditions in Study 2

Navon task			
Condition	RT (ms)		Acc (%)
Local Consistent	504.46 (66.83)		90.12 (9.08)
Global Consistent	550.37 (78.02)		90.81 (8.41)
The forced-choice triad task (80 triads)			
Thematic matching proportion mean (%)	Thematic-biased responder (thematic response > 48)	Taxonomic-biased responder (thematic response < 32)	Ambiguous responders
47.95 (15.40)	26 (23.0%)	36 (31.9%)	51(45.1%)
The semantic priming task			
Semantic priming type	RT (ms)		Acc (%)
Thematic	587.34 (74.09)		92.29 (10.82)
Taxonomic	626.06 (81.78)		91.15 (9.31)
Neutral	690.24 (98.32)		89.84 (7.76)

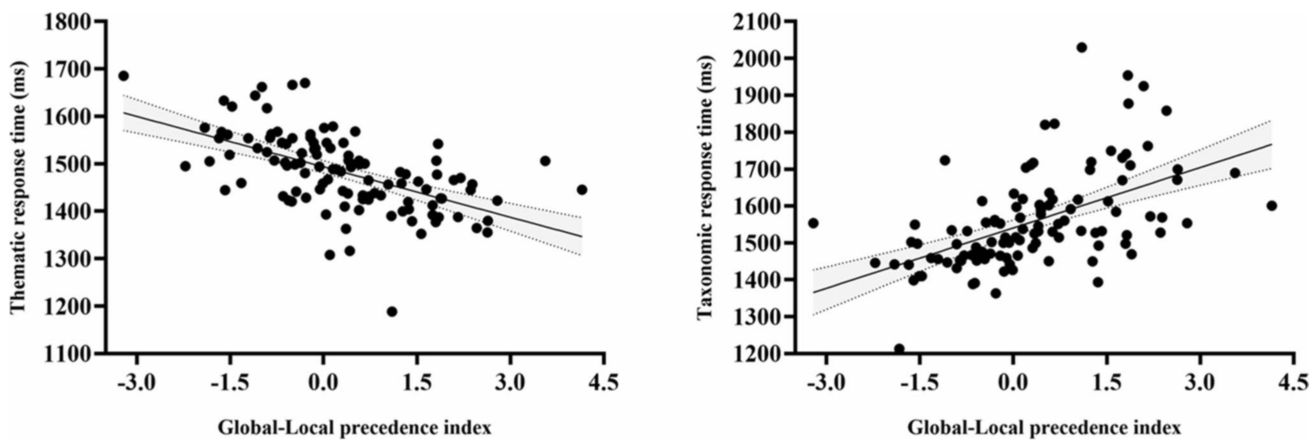

Fig. 2 Scatterplots: (Left) Relationships between the Global–Local precedence index and response speed for matching thematic-related options, (Right) and between the Global–Local precedence index and response speed for matching taxonomic-related options. Finally, the correlational analysis revealed that the Global-Local Precedence Index was significantly positively correlated with the proportion of thematic choices made by participants in the forced-choice triad task, $r = .444$, $p < .001$, 95% CI [0.282, 0.582] (see Fig. 3)

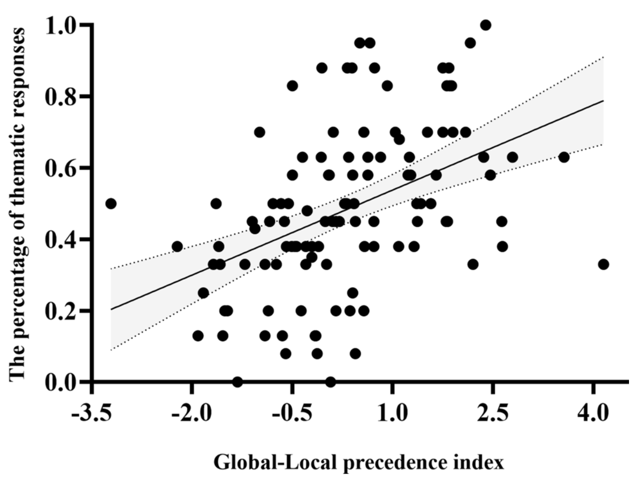

Fig. 3 Scatterplots: Relationships between the Global–Local precedence index and the proportion of thematic response

Second, as in Study 1, we calculated the average reaction time (ms) for correct responses under the congruent condition in the Navon letter task (see Table 2). A paired-sample *t* test revealed that participants responded significantly faster to Navon global letters ($M = 504.457$, $SD = 66.828$) than to Navon local letters ($M = 550.370$, $SD = 78.019$), $t(112) = -5.994860$, $p < .001$, 95% CI [−61.087, −30.738], Cohen's $d = -0.564$. Furthermore, following Study 1, we calculated the Global-Local Precedence Index for all participants. More importantly, correlation analysis revealed that the Global-Local Precedence Index was significantly negatively correlated with the reaction time for correct judgments of Gabor patch orientation under thematic priming, $r = -.525$, $p < .001$, 95% CI [−0.647, −0.377]; significantly positively correlated with the reaction time under taxonomic priming, $r = .360$, $p < .001$, 95% CI [0.187, 0.510], and not correlated

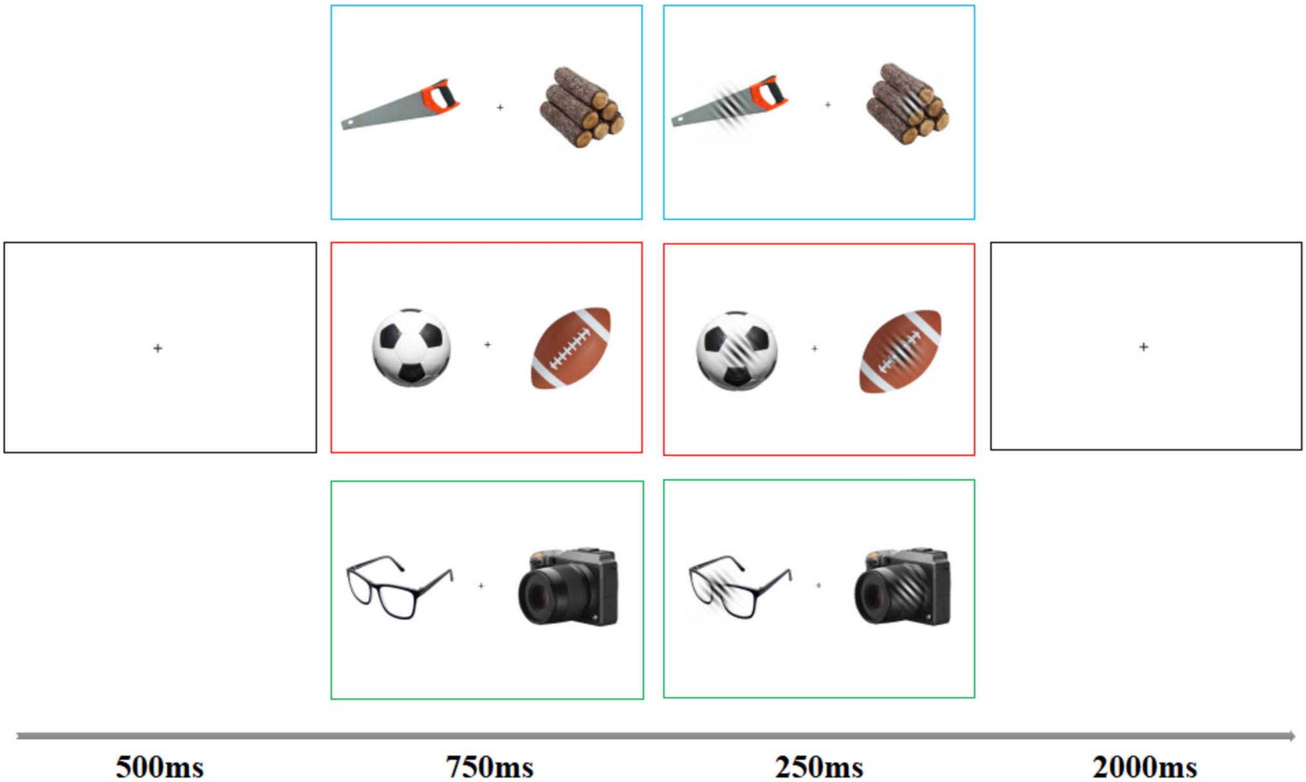

Fig. 4 The example of a semantic priming task

with reaction time under neutral priming, $r = .103$, $p = .278$ (see Fig. 5).

Finally, correlation analysis revealed that the Global-Local Precedence Index was significantly positively correlated with the proportion of thematic-related choices made by participants in the forced-choice triad task, $r = .532$, $p < .001$, 95% CI [0.385, 0.652] (see Fig. 6).

In summary, the findings once again demonstrate that individuals with higher levels of global precedence recognized thematic relations more quickly and exhibited a stronger preference for them.

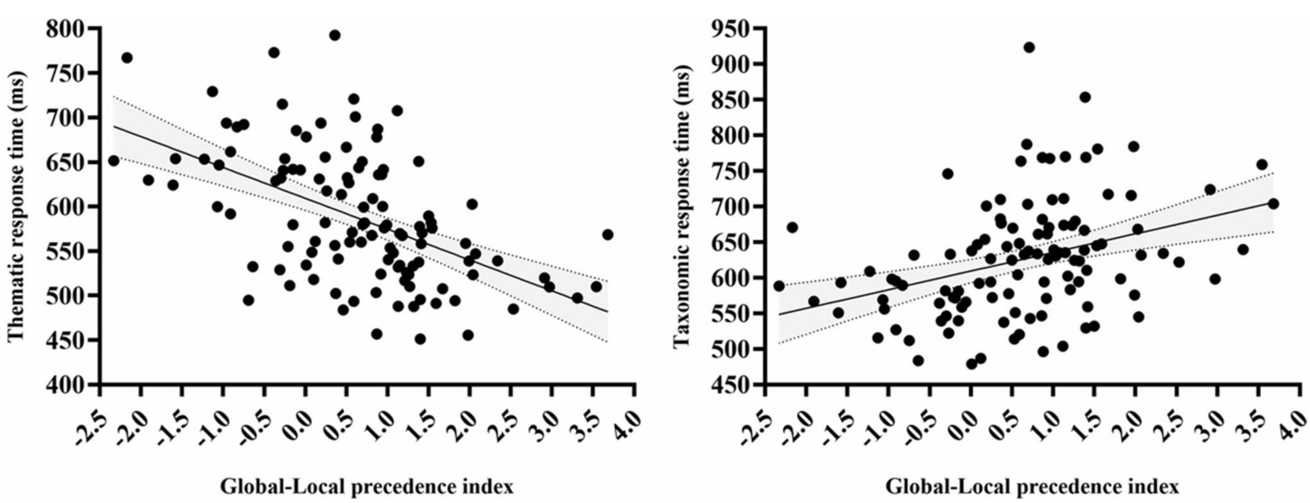

Fig. 5 Scatterplots: (Left) Relationships between the Global–Local precedence index and the average reaction times after thematic priming, (Right) between the Global–Local precedence index and the average reaction times after taxonomic priming

General discussion

This study investigated whether individuals with higher levels of global precedence recognized thematic relations faster and exhibited a stronger preference for them, through semantic processing tasks using words or pictures as stimuli. The study examined whether individuals with higher levels of global precedence recognized thematic relations faster and exhibited a stronger preference for them. Study 1 revealed that participants with higher Global–Local Precedence Index scores selected thematic options more frequently in the forced-choice triad task and identified thematic-related options more quickly in the similarity-matching task (a word-based task). Study 2 further demonstrated that participants with higher Global–Local Precedence Index scores continued to choose thematic options more frequently in a forced-choice triad task with a larger and more diverse set of stimuli. Additionally, these participants responded more quickly in correctly judging the orientation of Gabor patches following thematic priming in the semantic priming task (a picture-based task). These findings indicate that individuals with higher levels of global precedence recognize thematic relations more rapidly and exhibit a stronger preference for them, whether using a picture task or a word task.

This study provides more direct evidence that differences in global precedence may be one of the factors driving individual differences in thematic and taxonomic processing. Previous research has demonstrated that factors such as age, domain expertise, culture (East Asians and Euro-Americans), and levels of construal are closely related to individual differences in the relative strength of taxonomic and thematic relations (Estes et al., 2011; Mirman et al., 2017). Studies further suggest that these factors are closely associated with individual perceptual processing characteristics, particularly the global precedence effect (e.g., Li et al., 2024; Rezvani et al., 2020). Moreover, both behavioral experiments and neuropsychological studies have indirectly indicated that the representation of thematic relations may be closely tied to the perception of Gestalt consistency (Huberle et al., 2012; Lestou et al., 2014; Mirman et al., 2017; Maldei et al., 2020). This study found that individuals with higher levels of global precedence recognized thematic relations more quickly and exhibited a stronger preference for them. This finding provides more direct support for the notion that differences in global precedence may be one of the factors driving individual differences in thematic and taxonomic processing. However, it is important to acknowledge that this study primarily investigated the relationship between global precedence and the representation of established semantic links at the behavioral level. In addition, all triad tasks employed word triads, which may have somewhat limited the interpretation of the research findings. Future studies should consider incorporating both picture triads and word triads, and further explore the topic from additional perspectives, such as representational similarity in neuropsychological studies.

This study partially supports the dual hub account. The dual hub account (Mirman et al., 2017) posits a functional and neural dissociation between taxonomic and thematic relations. Previous research has suggested that differences in cognitive development, cultural and domain expertise, and level of construal can lead to distinct effects on thematic or taxonomic semantic processing, indicating that these relations are functionally dissociable (e.g., Estes et al., 2011; Mirman et al., 2017). This study further reveals that differences in individuals' perceptual processing characteristics also influence their performance in processing thematic and taxonomic semantics. This finding provides some support for the dual hub account.

Future research should further explore the relationship between local processing and the representation of taxonomic relations. In this study, we found that individuals with higher levels of global precedence recognized thematic relations more quickly and exhibited a stronger preference for them. This finding may also imply a close association between local processing and taxonomic relation representation (individuals who are more inclined toward local processing might be more sensitive to taxonomic relations). Moreover, the theoretical conceptual definitions of local processing and taxonomic relations (both emphasizing feature-based processing) align well with this speculation. However, we believe that caution must be exercised when drawing such inferences from the current study. First, this study examined the relationship between the global precedence effect and the processing of already-established semantic links. However, processing two concepts with

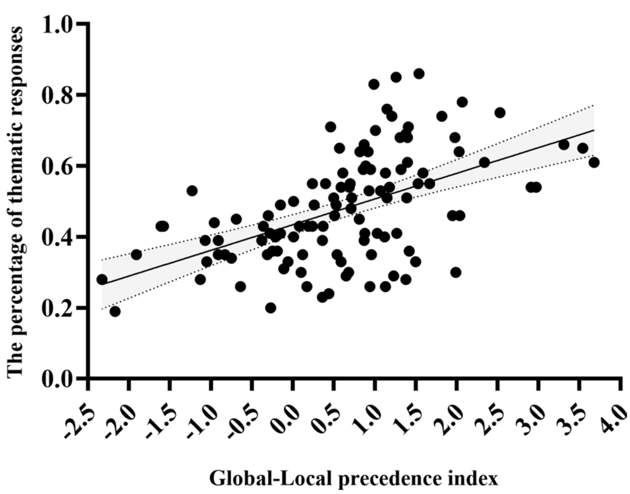

Fig. 6 Scatterplots: Relationships between the Global–Local precedence index and the proportion of thematic response

preestablished taxonomic similarities does not necessarily involve feature-based processing. For instance, noticing that American football and European football belong to the same semantic category may not require identifying detailed characteristics; it could simply result from knowledge imparted by teachers or parents (Maldei et al., 2020). Future research should explore this relationship further from the perspective of learning new semantic relationships. Second, although some studies suggest that taxonomically related concepts are likely to share visual features, which could increase fixation probability even if the images themselves do not share the similarity (Mirman & Graziano, 2012a, 2012b), the semantic priming task in Study 2 also required participants to judge the orientation of Gabor patches by recognizing visual features. When both tasks involve feature processing, it may lead to direct competition at the perceptual level, potentially slowing reaction times for judgments following taxonomic semantic priming. As a result, the association between local processing tendencies and the speed of recognizing taxonomic relations may also have been influenced. Future research should aim to mitigate this issue, for example, by applying Gaussian blur to objects involved in taxonomic relations to minimize perceptual feature interference.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.3758/s13423-025-02655-z.

Acknowledgements Thanks to Professor Joy J. Geng, Professor Garrett Honke, and Professor Christian Gerlach for their help with the experimental materials, which is very important for us to complete the research.

Authors' contributions Not applicable.

Funding This research is supported by the Northwestern Normal University's 2023 Graduate Research Grant Program (2023KYZZ-B035).

Data availability The data from this study have been uploaded to the OSF sharing platform at the following access link, and none of the experiments was preregistered.

https://osf.io/2yu9g/?view_only=c99bbd45354e4efe845b0985926feb27

Code availability Not applicable.

Declarations

Ethics approval The study was conducted after obtaining Institutional Review Board approval from the Department of Psychology at Northwest Normal University. We received the written consent of all participants before testing began. All procedures performed in studies involving human participants were by the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication All participants signed informed consent regarding publishing their data.

Conflict of interest The authors declared no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

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